

# DEV\_IMPL\_GUIDE - DD: ILM-CFG-01 — Customer Service Module

## 1. Purpose

This guide explains how to implement DD\_ILM-CFG-01-Customer\_Service-v1.0.md as executable SOPHIX behavior.

Use it with:

- DD\_ILM-CFG-01-Customer\_Service-v1.0.md
- DD\_SUB-WF-FRAMEWORK-01-Subscription\_Workflow\_Framework-v1.0.md when this DD participates in subscription workflow operations.
- DD\_SUB-WF-FRAMEWORK-01-WORKERS-v1.0.md when this DD references worker topics.

## 2. Runtime Actors

| Runtime component        | Responsibility                                                            |
|--------------------------|---------------------------------------------------------------------------|
| API pod / module service | Receives requests, validates, loads authoritative state, starts commands. |
| Owning module database   | Stores tables/configuration owned by this DD.                             |
| Reference modules        | Provide data through approved APIs/client wrappers.                       |
| Camunda                  | Runs BPMN process when this DD is workflow-driven.                        |
| Worker apps              | Execute logical worker topics.                                            |
| Drools                   | Evaluates packages when rules are declared.                               |
| Kafka/outbox             | Publishes success/failure and integration events.                         |

## 3. Before Implementation

Confirm these contracts from the DD:

| Contract                | Value                                                                                                                                                                                                                                                                                          |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Endpoint scan reference | DELETE /api/customer-accounts/{id}/flags/{flag_code}                                                                                                                                                                                                                                           |
| BPMN/process key        | configured BPMN process                                                                                                                                                                                                                                                                        |
| Drools package          | configured Drools package                                                                                                                                                                                                                                                                      |
| Tables                  | customer, customer_account, customer_account_flag, customer_account_flag_catalog, customer_account_notification_pref, customer_account_status_history, customer_contact_method, customer_interaction, customer_kyc_approval, customer_kyc_document, customer_note, customer_sub_status_catalog |
| Events                  | SubscriptionCreated, SubscriptionPaused, SubscriptionResumed, SubscriptionTerminated, WorkOrderCancelled, WorkOrderCreated, WorkOrderFinalized, WorkOrderPhaseFinalized                                                                                                                        |

## 4. Execution Flow

1. Apply migrations and seed configuration/catalog rows.
2. Implement the module service/repository layer around the tables and invariants.
3. Implement public/internal APIs from the DD.
4. Implement rule evaluation if the DD defines Drools packages.
5. Implement event production/consumption through outbox where the DD defines events.
6. Add integration tests for table constraints, API validation, and event payloads.

## 5. API And Command Handling

Endpoints found in the DD:

- DELETE /api/customer-accounts/{id}/flags/{flag\_code}
- DELETE /api/customer-accounts/{id}/subscription
- DELETE /api/customers/{id}
- GET /api/customer-account-flag-catalog?operatorCode=WIK
- GET /api/customer-accounts/search?key=flag&flagCode=NPD&value=true
- GET /api/customer-accounts/search?key=payment\_account\_number&value=<ref>
- GET /api/customer-accounts/{id}
- GET /api/customer-accounts/{id}/flags
- GET /api/customer-accounts/{id}/flags/{flag\_code}
- GET /api/customer-accounts/{id}/notification-pref
- GET /api/customer-accounts?customerId=...
- GET /api/customers/search
- GET /api/customers/search?key=msisdn&value=%2B254712345678&limit=10
- GET /api/customers/{id}
- GET /api/customers/{id}/contact-methods
- GET /api/customers/{id}/interactions?limit=20
- GET /api/customers/{id}/notes?limit=20&kind=...
- GET /api/customers/{id}/notification-prefs
- GET /api/subscriptions/subs\_01HX\_JANE\_TRIPLE\_PLAY
- GET /api/subscriptions/{subscription\_id}
- GET /api/work-orders/count?customerId=...&statusNotIn=COMPLETED, CANCELLED
- PATCH /api/contact-methods/{id}
- PATCH /api/customer-accounts/{id}
- PATCH /api/customers/cust\_01HX3K9M2P5Q8R1T4W7Y0Z3N6
- PATCH /api/customers/{id}
- PATCH /api/notes/{id}/supersede
- POST /api/contact-methods/{id}/verify
- POST /api/customer-accounts
- POST /api/customers
- POST /api/customers/cust\_01HX3K9M2P5Q8R1T4W7Y0Z3N6/kyc/final-approve
- POST /api/customers/cust\_01HX3K9M2P5Q8R1T4W7Y0Z3N6/kyc/l1-approve
- POST /api/customers/{customer\_id}/kyc/approvals
- POST /api/customers/{id}/contact-methods
- POST /api/customers/{id}/interactions
- POST /api/customers/{id}/kyc/final-approve
- POST /api/customers/{id}/kyc/l1-approve
- POST /api/customers/{id}/kyc/reject
- POST /api/customers/{id}/notes
- POST /engine-rest/external-task/fetchAndLock
- PUT /api/customer-accounts/acct\_01HX3K9M2P5Q8R1T4W7Y0Z3N7/sub-status

- PUT /api/customer-accounts/acct\_01HX3K9M2P5Q8R1T4W7Y0Z3N7/subscription
- PUT /api/customer-accounts/acct\_01HZP9Q8R7S6T5U4V3W2X1Y0Za/flags/FRAUD\_SUSPECTED
- PUT /api/customer-accounts/{id}/flags/{flag\_code}
- PUT /api/customer-accounts/{id}/notification-pref
- PUT /api/customer-accounts/{id}/sub-status
- PUT /api/customer-accounts/{id}/subscription

Implementation rules:

- Resolve actor user and role from auth context, not from public request body.
- Check idempotency before inserting a new operation or creating a side effect.
- Return the existing operation/result for the same idempotency key and same request hash.
- Return 409 IDEMPOTENCY\_KEY\_REUSED for the same key with a different request hash.
- Use transactions or unique constraints for concurrency-sensitive creation.

## 6. Data Loading

Load data in this order:

1. Authoritative aggregate state from the owning module.
2. Operator/module config from this DD's config tables.
3. Catalog/reference data from approved APIs or client wrappers.
4. Current in-flight operation state from the framework, where relevant.

Do not read current mutable state from cache for state-changing operations.

## 7. Drools Configuration

Packages found:

- No Drools package explicitly declared.

For each package:

- Define fact classes that mirror the request, current state, config, and reference data.
- Keep rule output as explicit decision fields.
- Unit-test accept and reject cases from the DD.
- BPMN should re-check critical rules after start if state may have changed.

## 8. BPMN And Worker Topics

Worker topics found:

- await-kyc-approval
- final-approve
- flag-evaluator-wik-v1
- flag-evaluator-yassn-v1
- kenya-kyc-leaders
- kenya-l1-kyc
- no-disturb
- sophix-superadmin

Implementation rules:

- Model service tasks with `camunda:type="external"` and the exact topic string.

- Worker handler must validate required variables before calling downstream services.
- Store outputs as process variables with stable names.
- Use message correlation keys exactly as specified by the DD.
- Add compensation paths for each external side effect.

## 9. Events

Events found:

- SubscriptionCreated
- SubscriptionPaused
- SubscriptionResumed
- SubscriptionTerminated
- WorkOrderCancelled
- WorkOrderCreated
- WorkOrderFinalized
- WorkOrderPhaseFinalized

Event rules:

- Emit success only after authoritative commit.
- Emit rejected/failure event only after compensation/final failure decision.
- Use the outbox table/dispatcher; do not publish directly inside the transaction boundary unless the platform standard allows it.

## 10. Smoke Tests

Add tests for:

- Happy path.
- Invalid role or trigger.
- Invalid state.
- Missing config/catalog row.
- Dependency failure.
- Retry/idempotency.
- Event payload correctness.
- Worker topic failure and compensation.

## 11. Common Mistakes

- Treating worker-topic names as shell commands.
- Querying another module's database directly.
- Reading current state from cache in a state-changing flow.
- Emitting success before final commit.
- Hiding manual recovery paths for partial external side effects.
- Creating one deployment per topic when a consolidated worker app would be easier to operate.